Failed changes and the Superset Problem: The importance of learning in models of syntactic change

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Abstract

Formal models of language change based on iterated learning have enjoyed considerable success in recent decades. One of the most influential of these accounts is the Variational Learner (Yang 2000, 2002). This model offers an attractive way of deriving patterns of diachronic change, particularly S-curve phenomena, from a probabilistic model of variation. However, we demonstrate that this theory makes additional problematic predictions: that failed changes are impossible, and that more expressive grammars should always be preferred over less expressive competitors. These problems are traceable to the reinforcement learning algorithm that Yang uses in his acquisition-driven model of change. After showing that both predictions are falsified by a wide range of examples from various languages, we motivate a solution using a variant of the Variational Learner built instead around hierarchical Bayesian learning, an approach widely used in recent cognitive science research. This result underlines the importance of careful attention to learning algorithms, and suggests that there may be limits on the explanatory potential of iterated learning models for patterns of grammatical change.

1 Introduction

Much research on language change has been focused on explaining the *S-curve generalization*: the observation that an innovation tends to start at a low rate of use, rise steeply, and then level off as it approaches categorical status (Aitchison 2012; Denison 2003; Kroch 1989; Labov 2001; Weinreich, Labov & Herzog 1968). Many models of language change have been constructed with the goal of deriving the S-curve pattern (Blythe & Croft 2012; Kauhanen 2017; Labov 2001; Reali & Griffiths 2010; Yang 2000, 2002).

The S-curve generalization entails a weaker claim, that language change is *directional*. That is, rates of use of an incoming variant tend to increase monotonically over time; rates of use of a declining variant tend to decrease monotonically over time.

In this paper, we engage with the model of directional change in general, and the S-curve phenomenon in particular, developed in Yang (2000, 2002). Yang's "Variational Learner" model is founded on the *grammar competition* framework of Kroch (1989), in which variation is treated as a sort of di- or polyglossia in which the competing systems happen to overlap to a substantial degree. The grammar competition framework implies that the learner's task is not to learn a single target grammar, the learning environment is grammatically heterogeneous. Instead, the learner uses her data to induce a probability distribution over multiple grammars that approximates the mixture of grammars in the learning environment.

According to the Variational Model, learning consists of assigning a weight to each competing grammar, proportional to its success in parsing sentences in the learner's input. This weight in turn determines the frequency with which the learner selects each grammar in her own productions. The link between parsing success and production is the core of Yang's account of directional change: if a grammar successfully parses a higher proportion of the input, it will be used to produce a greater proportion of the next generation's input, and so on.

The Variational Model is an elaboration of Andersen's (1973) model of change as driven by acquisition. For Andersen, each generation of language users produces a linguistic environment which feeds acquisition by the following generation. Learning may be a source of change, if the linguistic environment and the acquisition interact so that Generation $n+1$ constructs a representation of linguistic knowledge that differs from the representation of Generation n (see Figure 1).

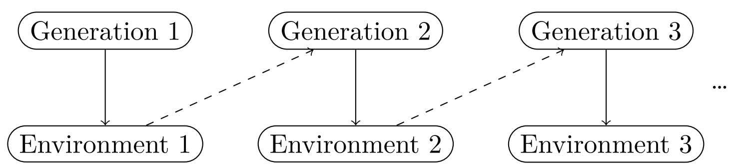

Figure 1: Acquisition-driven change. Solid arrows indicate causal relations of production: Generation n's linguistic knowledge produces linguistic environment n. This environment is used as evidence (dashed arrows) by Generation $n+1$ in the acquisition of their linguistic knowledge, which is then used to produce a new linguistic environment.

The broad family of acquisition-based approaches to S-curves and directed change contrasts with models which seek to derive the S-curve pattern from general facts about human social organization and interaction, while minimizing the explanatory burden placed on language-specific factors. The motivation for this second approach is that theorists interested in the diffusion of cultural and technological innovations had independently documented S-curve patterns in a wide variety of nonlinguistic domains (Rogers 1995; Ryan & Gross 1943; Tarde 1903). The generality of the phenomenon suggests that S-curves may commonly be found in language change, as a special case of S-curves in social change, because language is a social phenomenon. This perspective would be in line with the conclusions of linguists who have argued that the S-curve pattern is driven by social factors, notably social identification, that are only contingently related to the linguistic structures affected. In this paper, we do not systematically discuss social models of language change. Rather, we will focus on Yang's acquisition-based model for the bulk of the paper, motivating a revision and returning to discuss socially-driven models briefly in the conclusion.

Yang's Variational Model (henceforth VM) has a number of attractive features and has been very influential. Nevertheless, we will show that it encounters severe empirical difficulties. VM is unable to account for two well-attested historical patterns, which are sometimes overlooked as a consequence of the focus on S-curves: (1) the existence of failed changes, and (2) what we call the *Superset Problem*: the possibility of gradual loss of a linguistic feature without the rest of the grammar 'compensating', leaving a grammar which is strictly less expressive in terms of weak generative capacity. A well-known example of failed change is the rise and fall of affirmative *do* in Early Modern English. Examples of the Superset Problem include the loss of *pro*-drop between medieval and modern French, the loss of the subjunctive in English, and the slow loss of extraposition in several European languages (Wallenberg 2016). As we show in Section 2, VM predicts that failed change is impossible, and that superset grammars should always win in grammar competition (indeed, should go to probability 1 in a single generation). These demonstrably false predictions of the theory cast doubt on its correctness, and so call into question its derivation of directional change and the S-curve phenomenon.

However, these problems are due to a specific choice made by Yang (2000, 2002): the treatment of probability learning as a reinforcement learning problem, using the LINEAR REWARD–PENALTY (LRP) scheme (Bush & Mosteller 1951). The flaw in this theory of learning is that it rewards parsing success, but does not balance this goal against any other factors—notably, the goal of avoiding unwarranted expressivity, as motivated in work on the Subset Principle (Berwick 1985; Biberauer & Roberts 2009; Manzini & Wexler 1987). The problematic empirical predictions of VM—including directionality and the Superset Problem—are crucially dependent on this unforced theoretical choice.

We show that it is possible to maintain the basic structure of the Variational Model while avoiding problematic empirical predictions if we adopt a better theory of learning. While there are various options, we illustrate with a hierarchical Bayesian learner. This simple but powerful alternative learning model strives for a balance between empirical coverage (parsing success) and simplicity (a ceteris paribus preference for less expressive grammars). The result is a variant of Yang's theory that maintains both major framework-level commitments: grammar competition, and iterated learning as a mechanism of syntactic change. But it avoids the problematic empirical predictions, leaving room for failed changes and avoiding the Superset Problem.

The updated model makes dramatically different diachronic predictions. It no longer predicts that changes in progress should go to completion (deterministically or even by default), and it does not predict the S-curve phenomenon.

As we will discuss, though, it is not clear whether this is a bug or a feature: the empirical phenomena discussed in this paper show that directionality in syntactic change is not as robust as linguists have sometimes supposed, and not all changes follow the S-curve pattern. However, to the extent that these patterns are at least tendencies, they must receive an independent explanation, perhaps along the lines of the socially-oriented theories described above.

The paper is structured as follows. Section 2 reviews the structure of Yang's theory and key predictions. This section also shows how the use of LRP as a model of learning is crucially implicated in these predictions: there is no reason to assume that the same predictions would follow if any other model of learning were used in its place. Section 3 reviews case studies in grammatical change which exemplify the Superset Problem and failed changes, separately or together. Section 4 explains how a Bayesian variant of Yang's theory avoids the problematic predictions, and Section 5 concludes.

2 Competition, learning, and change

2.1 Competition

This section lays out the Variational Model (VM) of learning and change. The theory is highly modular, combining statistical learning with grammar competition and an acquisition-based model of grammatical change. Within this overarching framework, it is possible to "plug in" different models of grammar and learning and explore the consequences. This structural aspect of Yang (2002), and its importance for the diachronic consequences of the theory, has been largely overlooked in the subsequent literature. One intended contribution of this paper is to begin the program of exploring the range of possible theories within the broader VM framework.

VM treats change as driven by individual learning in a grammatically heterogeneous environment. An individual learns by updating a probability distribution over a set of grammars. The model is officially neutral as to what a grammar is. In Kroch (1989) and Yang (2002), a grammar is modeled as a specification of the values of a set of (typically binary) parameters. It will occasionally be useful in Section 3 to think of a grammar in the lexicalist sense, as a set of functional heads with associated feature specifications (Borer 1983; Chomsky 1995; Kroch 1994). In either case, a grammar is a discrete object that determines a strong generative capacity (the set of parsed strings that it can generate) and a weak generative capacity (the set of strings it can generate). A great virtue of grammar competition is that it reconciles a discrete conception of grammar with the well-attested fact that grammar change is gradual (*pace* Lightfoot 1979). Crucially, in these models an individual does not learn by selecting a single "target" grammar (as in Clark & Roberts 1993) but by inducing a distribution over grammars. This feature circumvents many problems identified in the parameter-setting literature.

There is currently no consensus about the nature of the set of discrete grammars that the learning process operates over (roughly, the "possible human language[s]" of Chomsky 1965, p. 25). This will be relevant at certain points in Section 3, where we bring real-world data to bear on the evaluation of the model: we cannot relate patterns of change directly to learning, but must first relate the patterns of behavior to a set of grammars, and then construe learning and change in terms of those grammars. This is important because no single grammar is taken to be responsible for the totality of any individual's linguistic behavior. As a result, it is itself a nontrivial analytical question how to map behavior (e.g., individual utterances) onto grammars. Nevertheless, in Section 3 we will be able to draw inferences about the set of possible grammars that are robust enough to allow us to link patterns of behavior to patterns of grammar learning and change.

2.2 Learning

For VM to make predictions about acquisition-driven patterns of historical change, two components must be specified: a theory of how the linguistic knowledge of each generation plays into the production of the associated linguistic environment (solid arrows in Figure 1), and a theory of how the next generation uses that environment to induce a probability distribution over grammars (dashed arrows). The latter component—the learning theory—has gotten the most attention in Yang's and subsequent work in the grammar competition/VM framework.

As a theory of probability learning, Yang (2000, 2002) proposes to use an influential algorithm from reinforcement learning, the *Linear Reward–Penalty* (LRP) scheme (Bush & Mosteller 1951). As Yang emphasizes, this algorithm is selected mainly for its simplicity, and many other choices are available in the literature. While the specific learning theory is perhaps not meant as an important commitment, the LRP scheme has become closely associated with the grammar competition framework in general and with Yang's VM in particular.

Yang does not compare the LRP model to any alternatives. However, we will show that the big predictions that Yang himself draws about language change all depend crucially on the LRP scheme, and that different learning theories lead to importantly different empirical predictions. A detailed comparative examination of the LRP scheme and other models of probability learning is thus sorely needed, along with empirical evidence that would allow us to compare them.

Bush & Mosteller (1951) proposed LRP as a simple learning model for reinforcement experiments in behaviorist psychology. An organism is presented with some set of choices—in the simplest case, between performing some action a and not performing it—and has some initial propensity $p \in [0, 1]$ for performing a in a fixed time period. In a classic Skinner box, the organism might be a rat, and the action might be pressing a lever. In each time period, the rat chooses to press the lever with probability p or not with probability $1-p$. If the rat presses the lever, it may be either rewarded (say, with food) or punished (say, with an electric shock). Similarly, it might be rewarded or punished if it fails to perform the action. According to the LRP scheme, when a reward is received contingent on some (non-)action, the probability of the rat performing that (non-)action in the next time step increases by some increment. Conversely, if a punishment is received the relevant probability decreases. The dynamics of the model can be expressed as follows, where p_t is the probability of the action at time t, p_{t+1} is the updated probability of that action, and $\alpha \in [0, 1]$ is a parameter determining the "learning rate".

$$p_{t+1} = p_t + \alpha(1 - p_t) \text{ if the action is rewarded} \tag{1}$$

$$p_{t+1} = p_t - \alpha(1 - p_t) \text{ if the action is punished} \tag{2}$$

Since the rat must choose one action or the other, these equations also determine the probability of the other action (not pressing the lever) at time $t+1$ as $1-p_{t+1}$. The intuition is that, each time action a is reinforced, it steals a little bit of probability from the other action. Similarly, when punished a gives a little bit of probability to the other action. With small α, the probability of a changes only a little bit after a reward or punishment. With large α, the rat might change its behavior wildly between different time steps depending on the pattern of reinforcement. Either way, the probability of an action tends toward 1 with repeated positive reinforcement, and tends toward 0 with repeated negative reinforcement. If the action is reinforced and punished in equal measure, its probability tends to remain fairly stable. These are the dynamics for the case of two possible actions. With three or more actions, the dynamics are the same except that the probability gained or lost by the relevant action has to be distributed over all of the other actions so that the total sums to 1.

Yang's proposal is to use LRP to model how children learn a probability distribution over grammars within the VM framework. Simplifying inessentially, assume that there are only two grammars, G^1 and G^2. A learner with no evidence should presumably the grammars equal initial weight, so at time t_0, $p_0^1 = p_0^2 = .5$. Yang assumes that each generation of learners encounters sentences drawn from a fixed linguistic environment E (produced by the previous generation). At each time step t, a sentence $S_t \in E$ is sampled and presented to the learner. This is analogous, in the LRP scheme's original domain of application, to the environment of the Skinner box at time t. The learner selects a grammar according to distribution p_t and attempts to use the selected grammar to parse S_t. This step is analogous to the organism selecting an action.

Suppose that the learner selects G^1. If G^1 can successfully parse S_t, then G^1 is reinforced, and p_{t+1}^1 increases relative to p_t^1.

$$p_{t+1}^1 = p_t^1 + \alpha(1 - p_t^1) \text{ if } G^1 \text{ can parse } S_t \tag{3}$$

$$p_{t+1}^1 = p_t - \alpha(1 - p_t^1) \text{ if } G^1 \text{ cannot parse } S_t \tag{4}$$

If G^2 is selected, the predictions are symmetric: parsing success is positively reinforced, and failure is negatively reinforced.

In each generation, a homogeneous population of learners is exposed to some number N of sentences drawn from the linguistic environment E. To simplify, assume that all sentences in E can be parsed by either G^1 or G^2, and that these grammars partially overlap in which sentences they can parse. Then the environment can be pictured as in Figure 2. a is the proportion of sentences

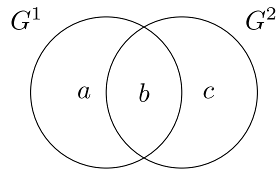

Figure 2: Weak generative capacity of G^1 and G^2 in relation to the linguistic environment E.

in E that can be parsed only by G^1, b is the proportion that can be parsed by either G^1 or G^2, and c is the proportion that can be parsed only by G^2. With probability b, the sentence presented will reward whichever grammar is chosen, and so on average it will not affect the dynamics. The dynamics of the system are completely determined by the quantities a and c—that is, the relative difference in probabilities of reinforcement and punishment for each grammar, given this linguistic environment—along with noise in the sampling process. If N is sufficiently large, then we can ignore sampling noise and the outcome of the learning process is well approximated by its behavior in the limit of infinite data. This is a simple function of the relative probabilities of reward and punishment for each grammar (see Bush & Mosteller 1951, p.316; Narendra & Thathachar 2012, p.163; Yang 2002, p.32).

$$p^1_\infty = \frac{a}{a+c} \tag{5}$$

$$p^2_\infty = \frac{c}{a+c} \tag{6}$$

It is worth noting a conceptual point about this model that will be important later. Reinforcement learning algorithms are designed to learn a probability distribution over ACTIONS—a “policy”, in RL jargon. As Sutton & Barto (2018) describe the field,

> Reinforcement learning is learning what to do—how to map situations to actions—so as to maximize a numerical reward signal. The learner is not told which actions to take, but instead must discover which actions yield the most reward by trying them. (p.1)

The success of a reinforcement learning algorithm does not, in general, require the learner to create a model of the world (indeed, this interpretation would be antithetical to LRP’s behaviorist origins). As a result, the distribution over grammars acquired by an LRP learner (p^1_t and p^2_t above) is the learner’s best guess about what to **do** in the next time step, in order to maximize the expected reward of being able to parse the next sentence presented. It would be inappropriate to think of this probability distribution as a theory of the underlying distribution of grammars in the world that was involved in the creation of the linguistic environment. Successful parsing is the only metric of success in the theory.

This conceptual point applies equally to other RL algorithms: success in a reinforcement learning scenario does not generally require the learner to acquire a causal understanding of the world. As we will show below, the orientation of Yang's LRP learner toward action selection, with a reward function determined entirely by parsing success, introduces systematic biases that make it a very poor learner of the underlying distribution of grammars in some cases. This property is the source of the pathological properties of LRP as a theory of probability learning for acquisition-driven theories of language change, and a crucial difference between typical RL methods and the more successful Bayesian model that we will introduce. The Bayesian learner differs in that it is properly construed as constructing a theory of the underlying distribution on grammars that is responsible for generating the observed data in the environment.

2.3 Change

In Yang's theory, the learning procedure just described feeds into language change as follows. Each generation acquires a probability distribution by applying LRP to the the environment generated by the previous generation. These now-mature language users then use the acquired distribution to produce a new linguistic environment for the following generation, and the process repeats. Yang's theory is silent on the details of how the linguistic knowledge of a generation of users produces the statistical properties of the linguistic environment. Surprisingly, we can learn a lot from this model without knowing how the distribution on grammars produces the linguistic environment. Yang assumes that the statistical properties of the sentences produced by any particular grammar are stable. In our two-grammar case, this means that the probability of each sentence type T in the linguistic environment produced by generation n can be computed from the statistical mixture of grammars that generation n is employing, and the probability that each grammar $G_{1/2}$ produces sentences of type T.

$$P(T) = P(T \mid G_1) \times P(G_1) + P(T \mid G_2) \times P(G_2)$$

Let T_1 be the type of sentences that can be produced only by G_1, T_2 be those that can be produced only by G_2, and $T_{1/2}$ be those that can be produced by either. In terms of Figure 2, this means that a is the probability that G_1 is selected, multiplied by the probability that G_1 would produce a sentence unparseable by G_2 (type T_1):

$$a = P(T_1 \mid G_1) \times P(G_1)$$
$$c = P(T_2 \mid G_2) \times P(G_2)$$

Again, we assume nothing about how the various grammars assign probabilities to these expression types.

As we saw above, the environment so derived is the input to the next generation's learning. The values of a and c in this environment are what controls the probabilities of G_1 and G_2 in the next generation's learning. Since there are only two options, all we need is the simple equation $p^1_\infty = a/(a+c)$, which depends on the distribution of sentences in the environment that are unparseable by each grammar. Using this, we can derive a simple formula that describes the condition under which the probability of a grammar can increase from one generation to the next. p^1_∞, the probability of G_1 for generation $n+1$, is greater than $P(G)$, the probability that G_1 had for generation n, if and only if the following holds:

$$\frac{p^1_\infty}{p^2_\infty} > \frac{P(G_1)}{P(G_2)} \text{ iff } \frac{\frac{a}{a+c}}{\frac{c}{a+c}} > \frac{P(G_1)}{P(G_2)} \tag{7}$$

$$\text{iff } \frac{a}{c} > \frac{P(G_1)}{P(G_2)} \tag{8}$$

$$\text{iff } \frac{P(T_1 \mid G_1) \times P(G_1)}{P(T_2 \mid G_2) \times P(G_2)} > \frac{P(G_1)}{P(G_2)} \tag{9}$$

$$\text{iff } \frac{P(T_1 \mid G_1)}{P(T_2 \mid G_2)} > 1 \tag{10}$$

$$\text{iff } P(T_1 \mid G_1) > P(T_2 \mid G_2) \tag{11}$$

In words, the probability of G_1 tends to rise if and only if G_1, when selected, is more likely to produce sentences of type T_1—those that are unparseable by G_2—than G_2 is to produce sentences unparseable by G_1. Note that this condition depends only on the conditional probabilities of each sentence type under the grammar that generates it, and is independent of the absolute probabilities of the relative grammars. An internal property of the competing grammars—their propensity to generate various sentence types—entirely determines which grammar emerges as the winner in grammar competition.

Figure 3 illustrates the predicted trajectory of change in Yang's model with a variety of starting points $p^1_0 \in \{0, .1, .2, \ldots, .9, 1\}$. This assumes that the "advantage" of G_1, i.e., the rate at which it produces T_1-sentences unparseable by G_2, is 0.25—only slightly greater than the corresponding rate for G_2, which produces G_1-unparseable sentences with probability 0.2. The only case in which G_1 does not overtake G_2 within 15 generations is when G_1 has initial probability 0. Otherwise, G_1 rapidly comes to have a probability indistinguishable from 1, even from a very low initial probability.

In Yang's model, the interaction of probability learning and generational transmission thus predicts two much-discussed properties of syntactic change noted in the Introduction: the directionality of change (specifically an S-shaped trajectory, as in Fig. 3), and a preference for more expressive grammars. However, the choice of the LRP scheme as a learning theory plays the starring role in these predictions. Specifically, when embedded into the broader picture of acquisition-driven change as a theory of learning, the LRP scheme predicts that changes in the probabilities assigned to grammars are driven entirely by inter-

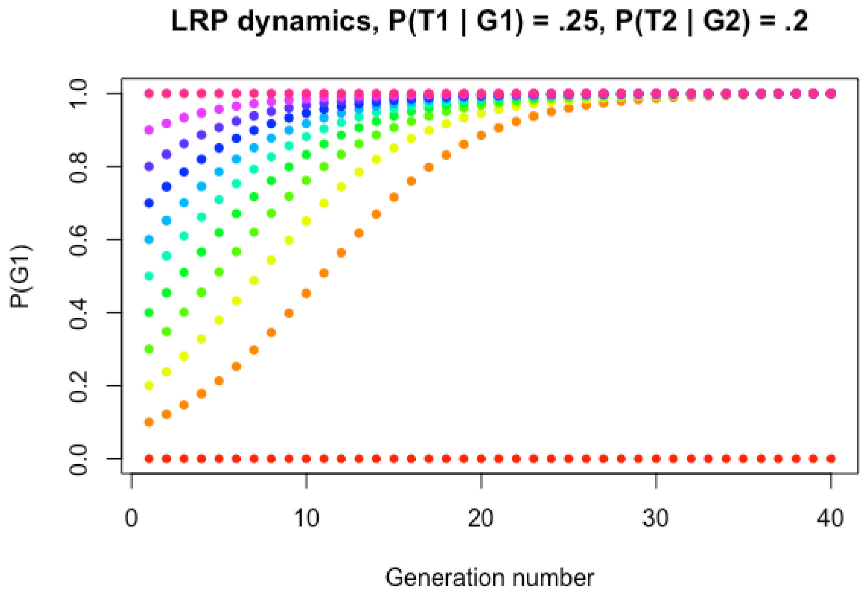

Figure 3: Generational dynamics of Yang's model. Even with a very slight parsing advantage, grammar G_1 rapidly goes to fixation (probability ≈ 1) for any initial probability greater than 0. Note the S-curve pattern of change apparent when the initial probability of G_1 is low.

nal properties of the grammars. The two key properties of grammars are their weak generative capacity, and whatever untheorized mechanism stochastically generates sentences from them.

> **The Crucial Property of LRP.** The probability of G_1 rises over generations, relative to G_2, **if and only if** G_1 is more likely, as an internal property of the grammar, to produce sentences unparseable by G_2 than vice versa.

Strikingly, this means that the question of whether G_1 is winning or losing in the competition depends **only** on an internal property of the grammars—how they produce sentence types. Assuming, as Yang does, that these propensities of grammars are stable, it follows that, if G_1 is winning in the competition at **any** point, it must win the competition at **all** points. Yang dubs this property of his theory the "Fundamental theorem of language change". Modified for the current formalism, it reads:

> **The fundamental theorem of language change:** G_1 overtakes G_2 if $P(T_1 \mid G_1) > P(T_2 \mid G_2)$.

For a "fundamental theorem", this prediction is strikingly dependent on contingent assumptions about how learning works. More crucially, it is empirically false: as we will show in some detail in Section 3, there are numerous cases in which one grammar G_1 has a clear parsing advantage over grammar G_2, and yet G_2 prevails over historical time. One set of examples involve a superset/subset relation among grammars. We call this the "Superset Problem":

> **The Superset Problem**. Suppose that two grammars G_1 and G_2 both have non-zero probability, and that the weak generative capacity of G_1 is strictly greater than that of G_2. As long as the linguistic environment contains sentences that are parseable only by G_1, the LRP-based theory predicts that G_1 must inevitably overtake G_2—and should do so in a single generation, as long as learners have sufficient data.

The Superset Problem follows directly from the LRP scheme. In general, if taking some action a_1 is rewarded whenever a_2 is, and also in some cases in which a_2 is not, LRP (quite sensibly) predicts that a_1 will be preferred. It is, after all, a guaranteed source of rewards that are *at least* as good, and sometimes better. By the same logic, if the superset grammar succeeds in parsing whenever the subset grammar does, and also in some cases in which the subset grammar fails, then the LRP scheme predicts that the probability of the superset grammar will go to 1 in a single generation. The situation is depicted in Figure 4. Since

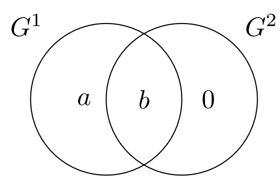

Figure 4: LRP predicts that superset grammars have an intrinsic advantage over strictly less powerful subset grammars, and should reach probability 1 after even a single generation.

there are no sentences generated by G_2 that G_1 cannot parse, $c = 0$ and we have simply

$$p^1_\infty = \frac{a}{a+c} = \frac{a}{a+0} = 1.$$

As an immediate consequence of the Superset Problem, LRP predicts that no grammatical feature can be lost over time in a way that is independent of the rest of the linguistic system. This is a very strong prediction, and—as we will see in a moment—subject to various counter-examples.

A second type of challenge to Yang's Fundamental Theorem comes from failed changes. Yang's theory predicts that a change, once initiated, must always proceed to completion. Yang points this fact out as a "Corollary" to his Fundamental Theorem (2002, p.132):

Once a grammar is on the rise, it is unstoppable.

Change in the relative probabilities of G_1 and G_2 is controlled entirely by stable, internal properties of the grammars. As a result, we have the following corollary to the corollary:

Failed changes are impossible.

Unfortunately, as we will see below, failed changes are not only possible but widely attested: they have simply not received much attention in the literature, perhaps because they are often considered less interesting than changes that go to completion, and/or because they are less likely to be noticed in the textual record.

3 Superset grammars and failed changes

In this section, we describe some case studies illustrating the Superset Problem and failed changes. Because of Yang's Fundamental Theorem and its Corollary, reproduced at the end of the previous section, any such cases are theoretically important, because they weigh against the LRP as a model of grammar learning: under such a model, these patterns cannot be explained in terms of the dynamics of a stable set of competing grammars.

More interestingly, in many of the cases discussed below, the failing grammar is a superset grammar, in that its weak generative capacity properly includes that of a competing grammar. We emphasize that weak generative capacity is the crucial point of comparison here, because the LRP algorithm rewards *any* successful parse of the input string.

These two issues, the Superset Problem and failed changes, are linked through their implications for the dynamics of Yang's theory of learning and change, but they can be dissociated. In the following subsections, we initially tackle them separately, before combining them. We show that a superset grammar can lose to a subset grammar (Section 3.1); that a change can fail intrinsically (Section 3.2); and finally that a change introducing a superset grammar can fail (Section 3.3).[1]

3.1 The Superset Problem: Subset grammars can outcompete superset grammars

Old French null subjects The grammar of Modern French differs from that of Old French in several interrelated ways (Adams 1987; Clark & Roberts 1993;

[1]This section includes corpus data from the following corpora, hereafter referred to by their acronyms: IcePaHC (Icelandic Parsed Historical Corpus, Wallenberg et al. 2011); MCVF (Modéliser le changement: les voies du français, Martineau et al. 2021); PLAEME (Parsed Linguistic Atlas of Early Middle English, Truswell et al. 2019); PPCME2 (Penn–Helsinki Parsed Corpus of Middle English, 2nd edition Kroch & Taylor 2000); and Beatrice Santorini's parsed corpus of Ellegård's examples, Santorini (2021).

Roberts 1993; Vance 1997; Yang 2002). Most relevant for our purposes, Old French had verb-second word orders (1) and allowed null subjects (2).

(1) A cel consel ot Nichodemus amis
at that meeting had Nichodemus friends
'Nichodemus had friends at that meeting.'
(*Roman du Graal*, Adams 1987, p. 4)

(2) Si firent grant joie la nuit
so made great joy the night
'So (they) had a great time that night.'
(Robert de Clari XII, Adams 1987, p. 2, translation added)

Based on data from Roberts (1993), Yang (2002) argues that French lost V2 in part because it was a null-subject language: once main clauses with null subjects are excluded, Middle French texts showed a greater frequency of distinctively non-V2 SXV orders than distinctively V2 XVS orders. We are interested in the next stage: the subsequent loss of null subjects. This is a case where a subset grammar outcompetes a superset grammar, because overt pronominal subjects have always been possible in French, in a range of orders, as in (3).

(3) a. Il en apelet e ses dux e ses cuntes
He of.it called and his dukes and his counts
'He called his dukes and his counts about it'
(*Chanson de Roland*, MCVF, 1100-ROLAND-MCVF-V,2.16)

b. De ço avun nus asez!
of this have we enough
'We have had enough of this!'
(*Chanson de Roland*, MCVF, 1100-ROLAND-MCVF-V,5.61)

In the 15th century, a grammar which could generate sentences with null and overt subjects was outcompeted by a grammar which could only generate sentences with overt subjects. In terms of weak generative capacity, this implies that a subset grammar can outcompete a superset grammar.

There are two immediate issues to discuss in this context. First, we must appreciate the crucial role of weak generative capacity. Typically, in European-style null subject languages, an overt pronominal subject is associated with marked interpretations, such as contrast. This means that examples like (3) may well not be assigned the same interpretation by a null subject grammar and a non-null subject grammar. But this is irrelevant to Yang's statistical learning algorithm, which rewards any successful parse.

The second issue is the following: is there a possible 15th-century French grammar, with null subjects and with no V2? That is, after the loss of V2, is it even possible for the superset grammar to exist? That is a theory-internal question, of course, and any answer will be contingent on a range of assumptions about syntactic theory. However, we believe that the answer must be yes.

The most basic reason for this is empirical: although the loss of null subjects is temporally very close to the loss of V2, there is general agreement that the loss of null subjects followed the loss of V2. In fact, Vance (1997) showed that the frequency of null subjects actually increased in the early 15th century, immediately prior to the loss of V2, and this plays an essential part in the accounts of the loss of V2 in Clark & Roberts (1993) and Yang (2002).

Another argument in support of the same conclusion concerns the account of the loss of null subjects in Clark & Roberts (1993). For Clark & Roberts, there are essentially two routes to licensing a null subject in Middle French: it can be governed by a finite verb in C, or licensed under Spec–Head agreement by a finite verb in I. When V2 was lost, the former route disappeared, but the latter route could in principle have remained. Clark & Roberts claim that it became impossible to license a null subject under Spec–Head agreement only when French "pronominal" rich agreement declined, an inflectional change which is dissociable from the loss of V2. There is therefore apparently no principled reason why a superset grammar, with no V2 and null subjects, could not have persisted. As a matter of fact, though, it didn't.

Decline of overt inverted-subject imperatives in Belfast English Henry (1995) describes two different dialects of Belfast English with respect to subject-verb inversion in imperatives. In the more restrictive Dialect A, inversion is limited to certain unaccusative verb classes, including telic verbs of motion (including cases where the telicity is determined by a particle or resultative predicate) and passives (once the general oddness of passive imperatives is controlled for).

(4) a. Go you there.
b. Run you away/home.
c. Be elected you president before the end of the year.
(Henry 1995, pp. 52, 55)

The more permissive Dialect B allows inversion with all verbs, including examples such as (5) which are impossible in Dialect A.

(5) Read you that book. (Henry 1995, p. 55)

There are other cases which imply that Dialects A and B stand in a subset–superset relation. For instance, in clauses with auxiliaries, with verbs compatible with inversion in Dialect A, the inverted subject must follow the main verb in Dialect A, but can also occur between auxiliary and main verb in Dialect B.

(6) a. Be you walking into the room just as she is leaving. (Dialect B only)

b. Be walking you into the room just as she is leaving. (Dialect A and B; Henry 1995, p. 56)

Henry (pp. 78–9) observes a change in progress whereby older speakers typically have Dialect B, whereas younger speakers typically have Dialect A. It appears, therefore, that a superset grammar is declining, at the expense of a subset grammar, even though in many cases, the input for acquisition of Dialect A will have been speakers of the more permissive Dialect B.

Henry discusses this phenomenon in terms which cannot be reconciled with Yang's model:

> "Thus, it seems that the task of the language learner is not to hypothesise or select, or even set the parameters corresponding to a grammar which can generate all the data in the input, rather it is to determine which grammar, from among the limited range made available by UG, can accommodate the majority of the data in the input. There may also be a preference for the simplest possible grammar …" (Henry 1995, p. 79)

Without going into details about the syntactic analysis that Henry assigns to imperatives in Dialects A and B, the crucial point is that a less expressive grammar is outcompeting a more expressive grammar, an example of the Superset Problem in progress.

3.2 Failed changes

Middle Dutch reflexives Recent theoretical interest in failed changes is largely due to Postma (2010). Postma's central example is the diachrony of Dutch reflexives. Earlier varieties of Dutch did not have distinct reflexive pronouns, but used simple pronouns like *hem* 'him' in reflexive contexts. Before the emergence of *sich* (later *zich*), the Modern Dutch simplex reflexive, a distinct form *sick* was briefly attested (in the Drenthe dialect which is Postma's focus, *sick* was concentrated in the 15th century).

Postma (2010) contains a lengthy discussion, based in L2 acquisition, of possible reasons why the *sick*-grammar failed and the *sich*-grammar spread. For the purposes of this paper, though, the crucial point is that the *sick*-grammar was on the rise for the first half of the 15th century, but (contrary to Yang's corollary), it was not unstoppable, but disappeared within a couple of generations.

Middle English ditransitives Bacovcin (2017) describes a failed change in the grammar of Middle English recipient–theme ditransitives. Verbs of transfer of possession, or future transfer of possession (Rappaport Hovav & Levin 2008), like Present-Day English *give*, select a recipient and a theme, while verbs of sending, like *send*, select a theme and a goal. Below, where we need a cover term for recipients and goals, we will use 'oblique'.

In Old English, the recipients in the former class were always marked with morphological dative case, while the goals in the latter class were marked with prepositional *to*. In Present-Day English, the two classes have largely collapsed, with both showing the well-known alternation between V–NP$_{\text{oblique}}$–NP$_{\text{theme}}$ and V NP$_{\text{theme}}$–*to* NP$_{\text{oblique}}$ orders.

The Old English system collapsed as morphological dative case disappeared in early Middle English, a process that was complete by c.1300, except for in the more linguistically conservative southeast of England. The result of the collapse was a transient state in which both oblique–theme and theme–oblique orders were possible, with the oblique argument either marked by *to* or unmarked in either order. The ditransitive structures which were grammatical during this transitional period are a proper superset of those which are grammatical in Late Middle and Modern English.

(7) a. hamot ʒet þurch hire forbisne & þurch hire hali beoden
she.may yet through her example and through her holy prayers
ʒeoue strencðe oðere
give strength others
'She may yet, through her example and through her holy prayers, give strength to others.'
(V–DO–IO, *Ancrene Riwle*, PPCME2, cmancriw-1,II.113.1414)

b. ne talde ha þen engel nan tale.
NEG told she the.DAT angel no tale
'She did not tell the angel any tale.'
(V–IO–DO, *Ancrene Riwle*, PPCME2, cmancriw-1,II.54.521)

c. & wið þe gode iosaphat sendeð beoden to sondesmon
and with the good Jehoshaphat send prayers to messenger
'And with the good Jehoshaphat, send prayers to a messenger.'
(V–DO–PP, *Ancrene Riwle*, PPCME2, cmancriw-1,II.193.2747)

d. þu leauedest to oðre men alle richesces
you left to other men all riches
'You left all riches to other men.'
(V–PP–DO, *Ancrene Riwle*, PPCME2, cmancriw-1,II.130.1698)

Although it initially appears that the maximally permissive behavior illustrated in (7) reflects a superset grammar, this is in fact controversial. For Bacovcin, the patterns observable in the transitional period are analyzable as a product of three distinct grammars, none of them strictly more expressive than the eventually successful grammar. Below, we add a fourth grammar (the OE grammar) for completeness.

1. The OE grammar: recipients marked with dative, goals with *to*, regardless of order;

2. The PDE grammar: V–$NP_{oblique}$–NP_{theme} / V–NP_{theme}–*to* $NP_{oblique}$;
3. An all-bare grammar: no marking of obliques, in either order;
4. An all-*to* grammar: *to*-marking of obliques, in either order.

Clearly, this enumeration of grammars is a theoretical commitment, and alternatives are easy to imagine. However, under any analysis that we can envisage, at least one grammar generating some of the 'transient' orders in (7) must have begun to spread, and then failed.

3.3 The Superset Problem in a failed change

English affirmative declarative *do* The best-known failed change is one where the failing grammar is also a superset grammar. This is the case of English affirmative declarative *do*. *Do*-support in Present Day English is a morphosyntactic phenomenon where the morpheme *do* is inserted as a last resort in finite clauses to host verbal inflection. This means that *do*-support occurs in environments in which two other mechanisms fail.

The first of these is affix-hopping (Chomsky 1957): when V and T are structurally local (more precisely, when no head intervenes), the verbal inflection on T lowers to V postsyntactically (Embick & Noyer 2001).[2]

(8) The dog ∅ bark-s.

The second mechanism is raising of an auxiliary to T. In this case, the auxiliary can host the verbal inflection.

(9) The dog ha-s *t* barked.

When both of those mechanisms are unavailable, *do* steps in. There are two main cases where this occurs. The first is when some other head intervenes between V and T, most commonly negation.

(10) The dog do-es n't bark.

The other is when T raises to C, for instance in matrix questions, leading to a surface configuration in which verbal inflection is not local to the verb (the TP projection intervenes)

(11) Do-es the dog t_{do} bark?

Note that this presentation of *do*-support assumes that *do* is a 'dummy element' syntactically and semantically, inserted as a last resort. In Middle English, in contrast, *do* was semantically contentful, used as a causative verb (although it

[2]Alternatively, abstract inflectional features on T agree with the verb subject to a similar locality constraint.

also had some expletive-like uses, for instance in VP-ellipsis). So there arguably was no such dummy element in Middle English. However, there was also no need for a dummy element, as Middle English grammar included generalized V-to-T raising, so the verbal inflection in T always had a host.

Do-support is relevant to failed changes because of an intermediate grammar, identified by Ellegård (1953) and investigated in detail by Ecay (2015). A hallmark of this grammar is that *do* occurs in affirmative declarative clauses, outside emphatic and contrastive uses.[3] An example is in (12).

(12) and at mas in presence of cristes body he doth kis hir
(Ellegård 1953, p. 191)

Affirmative declarative *do* emerged at the start of the Early Modern English period and peaked around 1575. It was never close to categorical: Ecay (2015) gives the maximum rate of use as $< 10\%$. Affirmative declarative *do* is therefore a failed change.

Of more interest is the way that this transient use of *do* relates to the apparently more stable Middle English and Present-Day English grammars of *do*. Ecay analyses affirmative declarative *do* as indicative of a use of *do* as a v head, marking the presence of an external argument, or perhaps specifically an agent. This lexical entry for *do* is an addition to other contemporaneous *do*s: *do* was apparently already in use as a dummy element by this point, for instance (this point was made cogently in Kroch 1989, which demonstrated a constant rate effect holding between the emergence of *do*-support of the modern type, and the loss of V-to-T raising, up until the peak of agentive *do* around 1575).[4] So the transient grammar, with *do* as a v head, is apparently strictly more expressive than other grammars which lack this category for *do*. In other words, the failed change plausibly involved the failure of a superset grammar to spread.

To evaluate this claim, we need to return to two themes mentioned repeatedly above. The first is to insist once again that the operative notion of expressivity for discussing Yang's VM is weak generative capacity. The transient *do* in affirmative declarative clauses is structurally and semantically distinct from Middle English and Present-Day English *do*s, but that is beside the point. The essential point is simply that *do* in certain affirmative declarative clauses rose, and then fell.

The second point again concerns the enumeration of grammars. For the failing grammar to be a superset grammar, we need to convince ourselves that a grammar can contain not only the little v head *do* described by Ecay, but also the last-resort dummy *do* in evidence in Present-Day *do*-support. We have no evidence to support or refute this possibility, but we can offer the following perspective. The Early Modern English context was one which included a dummy

[3] It is probably impossible to identify which uses of *do* are emphatic in the textual record. However, there is no reason to believe that the rate of emphasis has changed over time, so changes in the rate of *do* in affirmative declarative clauses should be treated as independent of emphasis.

[4] Although see Ecay (2015) for critical discussion.

element *do* and a *v* head *do*. We assume that a grammar, in the relevant sense, is just a set of lexical items with specification of their properties (loosely as in Borer 1983). So we would expect that there is a grammar containing the two *do*s just described, unless some principle can be adduced to block such a grammar. We are unaware of such a principle, so we expect that that grammar exists. That grammar would be a superset grammar relative to Present-Day English grammars which contain only the dummy *do*, and that superset grammar failed to spread.

Icelandic headed *wh*-relatives At the start of its written history, Icelandic typically marked headed relative clauses with the complementizer *er*.

(13)

in	helga	María,	er	bar	Drottin
the	holy	Mary	REL	bore	Lord

'the holy Mary, who bore the Lord'
(IcePaHC, 1150.homiliubok.rel-ser,.120, Sapp 2019, e6)

Over the course of this history, it has gradually replaced *er* with another complementizer, *sem*, in these contexts.

(14)

Þetta	er	maðurinn	sem	María	hitti	í gær.
this	is	man-the	that	Mary	met	yesterday

'This is the man that Mary met yesterday.' (Thráinsson 2007, p. 407)

However, Brendel (2023), Sapp (2019), and Youmerski (2017) show that between roughly the 15th and 18th centuries, headed relatives briefly assumed a range of other forms. Most notably, interrogative forms could be used as relativizers.

(15)

kom	einn	kóngur	virðuligur	og	voldugur	hver	er	hét
came	a	king	honorable	and	mighty	who	REL	was.called

Translatíus
Translatius

'There came an honorable and mighty king who was called Translatius'
(IcePaHC, 1450.ectorssaga.nar-sag,.54, Sapp 2019, e6)

This alternative relative clause structure was never close to being a primary relativization strategy—those were always *er* and *sem*, possibly supplemented by demonstrative *sá* (Sapp 2019). Rather, it was co-opted as a secondary strategy for a few centuries, and then disappeared.

It is plausible that *wh*-relatives were introduced as a result of contact, with Latin and with other Germanic varieties (Brendel 2023; Youmerski 2017). That is orthogonal to our main point, though: they became a small part of the grammar of Icelandic, persisted for centuries, and then disappeared. The superset grammar, containing interrogative and demonstrative forms which could be used in headed relatives, failed.

Almost a superset grammar: The CM grammar Truswell (2021) documents an unusual grammar, which he calls the 'CM grammar', most evident in the Edinburgh manuscript of *Cursor Mundi*, a mid-14th century northern Middle English text. The grammar is characterized by verb-raising to a position preceded by two A′-positions in matrix clauses. In embedded clauses, a complementizer makes one of the two positions unavailable, but the other remains. Accordingly, the CM grammar generates matrix verb-third and embedded verb-second orders, as in (16).

(16) a. Of þis t^{o}wþe hard es t^{o}wþe to find
of this truth hard is truth to find
'Of this truth, truth is hard to find.'
(PLAEME, edincmat.180, Truswell 2021, p. 13)

b. Men wat þat fu[l] ner es som comād
men know that full near is summer coming
'Men know that summer is drawing near.'
(PLAEME, edincmbt.258, Truswell 2021, p. 12)

Truswell argues that this grammar can also generate matrix verb-second examples such as (17), either because one of the two initial A′-positions can remain empty or because a single constituent can move from one position to the other.

(17) Ful g^{a}iþeli was tat g^{a}ce te g^{a}nt
full readily was that grace thee granted
'That grace was readily granted to thee.'
(PLAEME, edincmat.1057, Truswell 2021, p. 9)

With respect to matrix clauses, then, the CM grammar is strictly more expressive than the dominant English grammars of the time, which generate V2 orders but only a restricted set of V3 orders. With respect to embedded clauses, the situation is more subtle: there is no strict inclusion relationship between between the weak generative capacities of the two grammars. The dominant Middle English V2 grammars do not generate embedded verb-second orders with inversion like (16b). However, they can generate verb-late orders like (18), which cannot be generated by the embedded-V2 CM grammar.

(18) Sey me sodt p charite þ^{t} þu þerofen ne Lye
say me sooth by charity that thou thereof neg lie
'Tell me truthfully, out of charity, that you are not lying about that.'
(PLAEME, adde6at.88)

The CM grammar is not strictly a superset grammar, then. However, Truswell demonstrates that it had a clear quantitative parsing advantage over other Middle English grammars which have been identified: for almost every known Middle English text, the CM grammar can parse a greater number of sentence tokens than other Middle English grammars.[5]

Yang's model therefore predicts that the CM grammar should outcompete all other English grammars. However, this did not happen: positive evidence for the CM grammar is very short-lived, covering barely more than a century. Even *Cursor Mundi*, the text in which the grammar is most evident, shows only scattered examples which are unambiguously indicative of the CM grammar, as opposed to conventional ME V2 grammars.

This might seem to suggest that the distinctive word orders associated with the CM grammar are simply noise: idiosyncratic, strictly ungrammatical tokens, perhaps produced as a way of satisfying the metrical constraints of the text, perhaps produced as a result of over-literal translation from French sources. However, we do not believe that the distinctive word orders in (16) can be explained away like this. These word orders reveal structured heterogeneity, in that they are associated with northern texts around the 14th century. If these orders were noise, we would not expect to see this kind of conditioning of their distribution.

We therefore conclude that the CM grammar, while not strictly a superset grammar, is a grammar which had a clear parsing advantage over other ME grammars and yet failed, within roughly a century. The change which introduced the CM grammar (regardless of whether it had an endogenous or contact-based origin), failed.

3.4 Discussion

The list of documented failed changes, and/or declining superset grammars, is quite long. As well as the cases above, it includes at least:

- Loss of null subjects in Brazilian Portuguese (Adams 1987);
- Rise and fall of auxiliary ellipsis in Early Modern German (Breitbarth 2005);
- Decline of subjunctive *be* in English (Kastronic & Poplack 2021);
- Decline of extraposition across French, Brazilian Portuguese, Icelandic, and English (Wallenberg 2016).

This presents a robust, widespread challenge to the model of learning and change proposed in Yang (2002).

The question is how to respond to the challenge, taking into account the structure of Yang's model. A challenge like this should absolutely not be taken

[5]No restrictive grammar which has been explicitly described by any researcher has a 100% rate of successful parsing across any Early Middle English text. This is not surprising given the widely acknowledged syntactic heterogeneity of Early Middle English. We assume, in keeping with Kroch's general perspective on syntactic variation and change, that this is in fact the norm: any individual commands multiple competing and partly incommensurate grammars, and their heterogeneous linguistic behavior reflects this polyglossia.

as an invitation to reject Yang's theory wholesale, but rather to identify the components of the model which underpin the problematic predictions, and explore alternatives. In the rest of this section, we briefly discuss three such components. Each of these would reward further investigation, but the third choice point, the learning algorithm, offers the clearest immediate prospect for improvement. We discuss an alternative learning model in Section 4.

Which grammars are competing? The first choice point concerns the nature of the grammars which are in competition. In order to know when a subset grammar has outcompeted a superset grammar, we have to be able to recognize a grammar. However, as Figure 1 illustrates, we don't perceive grammars directly, but rather samples of the strings that grammars can generate.

To investigate grammar competition, we therefore must be able to unambiguously pair observable strings with members of the class of competing grammars. However, the identification of competing grammars is itself a theoretically loaded issue, as syntactic theory has not yet converged on a well-defined characterization of "possible grammar".

One possible response to the Superset Problem would then simply be to deny that the superset grammars are available to the learner. We have alluded to this with respect to two case studies above: Middle English ditransitives and Early Modern English affirmative declarative *do*. However, a clearer illustration of this possibility comes from the discussion of extraposition in Wallenberg (2016). Wallenberg describes a long-term decline in the frequency of relative clause extraposition, across English, Icelandic, French, and Portuguese. Sentences like (19a) are declining in frequency, at the expense of examples like (19b).

(19) a. Another method … may be employed which is less open to the above objection.

b. Another method which is less open to the above objection may be employed. (Wallenberg 2016, e237)

This would appear to be a clear example of the decline of a superset grammar (generating extraposed and in situ orders) at the expense of a subset grammar (generating only in situ orders). However, Wallenberg argues that the extraposed and in situ variants are products of two distinct competing grammars: for Wallenberg, there is no one grammar which generates both variants, but rather one grammar which categorically produces extraposed relatives and one grammar which categorically produces in situ relatives. If that is the case, there is in fact no subset–superset relationship between the two relevant grammars.

Wallenberg's ingenious claim contrasts with the mainstream position (Biberauer & Roberts 2009; Roberts 2022), in which a single grammar can generate optionality. We take Henry's discussion of subject inversion in Belfast English imperatives (see above) to demonstrate that two grammars can stand in a subset–superset relation with respect to weak generative capacity.

Henry argues that in Dialect A, the verb remains in V, while in Dialect B it raises to C. In Dialect B, the subject can either remain in situ within VP or raise to the canonical subject position (Spec,AgrSP in Henry's terms). The result is generalized inversion in Dialect B, as a consequence of V-to-C movement, but inversion in Dialect A only in the case of an unaccusative subject, in situ in a postverbal internal argument position.[6]

If this is correct, then Dialect A and Dialect B are differentiated by more than an extra optional operation: the verb obligatorily surfaces in different positions in the two grammars. This is significant in the context of Wallenberg's approach to optionality: because the 'superset' nature of Dialect B does not reduce to a single extra optional rule, it is not possible to factor out Dialect B into two distinct grammars, in the way that Wallenberg proposes for extraposition.

A more interesting Wallenbergian approach to the Belfast imperative data would be to propose that Henry's analysis in fact reflects *four* competing grammars, each with a single obligatory verb position and a single obligatory subject position:

1. Verb in V, subject remains within VP (specifier or complement, depending on external/internal argument status);

2. Verb in V, subject in Spec,AgrSP;

3. Verb in C, subject remains within VP;

4. Verb in C, subject in Spec,AgrSP.

Henry's Dialect A corresponds to grammars 1–2 (grammar 1 generating limited inversion with unaccusatives; grammar 2 generating uninverted orders). Dialect B corresponds to grammars 3–4 (both generating inversion in the simplest cases; grammar 3 being responsible for orders like (6b) and grammar 4 being responsible for (6a)). None of these four grammars stands in a subset–superset relationship with any of the others, as the reader can verify, which in principle is sufficient to permit the extension of Wallenberg's approach to the Belfast imperative data.

However, this is incompatible with Henry's description of the empirical facts. Henry's analysis implies that there are two grammars competing in Belfast English, and not four. The reason is that speakers of Dialect B, for instance, accept the orders generated by grammars 3 and 4 equally. This should not be the case if these orders reflected the output of two independent competing grammars: speakers should be free to assign probabilities to each grammar independently, for instance assigning a high probability to grammar 3 and a low probability to grammar 4; perhaps assigning a high probabiliy to grammars 1 and 3 and a low probability to grammars 2 and 4, etc. This is not what Henry

[6] There is probably more to say about the syntax of uninverted imperatives in Belfast English. We assume that uninverted examples like *Everyone go home*, if grammatical, reflect the presence of a distinct grammar. The reason for this assumption is that Henry's analysis of Dialect B does not appear to be compatible with such examples, and if a separate grammar were needed to account for such examples under Dialect B, it is natural to assume that it is also available to speakers of Dialect A. However, we do not believe that this affects the argument that we are developing here.

describes. Rather, Henry describes a clustering whereby some speakers have the verb in C, with variation in the subject position (corresponding to grammars 3 and 4), while others have the verb in V, again with variation in the subject position (corresponding to grammars 1 and 2). If this is an accurate description of the envelope of variation in Belfast, then we cannot extend Wallenberg's approach (which implicitly rejects such a clustering) to capture the decline of the superset grammar. This leads us to conclude that it is indeed possible for a single subset grammar to outcompete a single superset grammar.

What gets rewarded? A second possible response to the Superset Problem would be to revise the assumption in Yang's model that the criterion for success is any successful parse of the input. This is what drives our focus on weak generative capacity in the discussion in this section: the criterion of parsing success entails that it doesn't matter what structure a grammar assigns to the input, so long as it can parse that structure, somehow. This assumption has consequences for many of the case studies discussed in this section. For instance, both dialects in Henry's study of Belfast English imperatives can parse the sentence *Go you away*, but they assign different structures to the sentence, with *Go* in V for Dialect A but in C for Dialect B. There is therefore no subset–superset relationship between the grammars in terms of strong generative capacity.

With this consideration in mind, it would be natural to explore the possibility that Yang's model should be revised to make reference to strong, rather than weak, generative capacity. This would involve revising the algorithm presented in Section 2.2 as follows. First, rather than conceiving of the linguistic environment E as a set of sentence strings, E would be a set of structured sentences. The learning model would then consist of sampling a structured sentence $S_t \in E$, rewarding a grammar G^1 (according to equation 3) if it can generate that structured sentence, and penalizing G^1 (according to equation 4) if it cannot generate that structured sentence.

However, we believe that this is less realistic than the model as presented in Section 2.2, which relates strictly to weak generative capacity. The reason relates to the separation between I-language and E-language implicit in Figure 1. Learners do not have direct access to the syntactic structure of tokens in their input—if they did, the acquisition process would be significantly simplified. It would therefore be putting the cart before the horse to define parsing success in terms of strong generative capacity.

Nevertheless, there is a clear sense in which the criterion based on parsing success is insufficient. In everyday communication, it is not enough to successfully parse a string. Rather, an audience has to interpret the string in line with the speaker's communicative intention—or at least believe that they do. This is relevant to the issue of weak vs. strong generative capacity because the compositional interpretation of a sentence is determined by the sentence's syntactic structure, rather than just the unstructured sentence string. This might lead one to propose that the criterion for rewarding a grammar should be not successful parsing of a string, but successful communication: a grammar is rewarded if it can parse the input in a way which leads to an interpretation in line with the speaker's communicative intention, and otherwise penalized.

Although there is some intuitive sense in which such an approach would improve over the learning algorithm as presented by Yang, there are several problems which we believe cannot be satisfactorily resolved at present. First, there is the obvious matter (Sperber & Wilson 1986) that the learner does not know the speaker's communicative intentions, and an utterance does not transparently encode those intentions. Secondly, misunderstanding is widespread, and does not uniformly impede communication, so failure to infer communicative intention accurately should not be uniformly punished. Third, ambiguity is widespread, and recognition of one interpretation of a structurally ambiguous sentence may in principle reflect only partial success. Finally, grammar is far from the only module involved in inferring communicative intentions, and pragmatic capacities are still developing while syntactic acquisition is in progress. It would therefore be unrealistic to expect the inferred communicative intentions to match the speaker's actual communicative intentions, even in a case of perfect syntactic acquisition.

Faced with all of these challenges, we do not believe that a sufficient response to the problems identified here will involve *merely* replacing the reward criterion of parsing success with a criterion based on successful recognition of (e.g.) communicative intentions. However, such a modification is desirable in its own right, and could in principle be integrated with the Bayesian learner that we describe in the next section (see, for instance, Frank, Goodman & Tenenbaum 2009; Xu & Tenenbaum 2007).

4 Grammar competition with a hierarchical Bayesian learning model

We have identified two problems for a theory of syntactic change based on the Variational Model and using the LRP algorithm: the Superset Problem and failed changes. There are no immediate prospects for addressing these problems by varying other components of the VM. This leads us to focus on the choice of the LRP algorithm itself. As noted in section 2.3, this choice is central to the diachronic predictions in Yang's theory.

The LRP algorithm is problematic because it cares only about successful action choice, here cashed out as parsing success. A model of syntactic change based in grammar competition needs to accommodate multiple, possibly conflicting and interacting, factors leading to the kind of nonmonotonic dynamics seen in failed changes. In this section, we propose a variant of the VM which solves these problems by substituting a Bayesian learning theory for the LRP scheme.

A Bayesian learner is not the only one that can avoid the problems noted for Yang's LRP, though it is attractive in that it has had much success in recent cognitive science. However, the crucial feature of the Bayesian approach, when embedded in the VM framework, is that it balances the need for a grammar to be able to parse the sentences presented against a built-in preference for grammars that are as restrictive as possible. By contrast, the LRP model that Yang adopts considers only the first factor, and so tends to prefer less restrictive grammars, leading to the Superset Problem.

4.1 Background on Bayesian learning

The need to balance between restrictiveness and data coverage is closely related to the statistical tradeoff between formulating models that can account for the evidence received (fitting the data), while avoiding models that are so powerful that they can account for anything (overfitting). Modeling grammar learning as Bayesian inference leads automatically to a balance between these two pressures. This is because Bayesian learning automatically balances a model's ability to fit the data against an intrinsic preference for simpler models. As we will see, this approach avoids the problems noted above for the LRP scheme. In particular, the Superset Problem does not arise: a Bayesian learner will prefers simpler grammars unless a more powerful grammar is required in order to account for the data. Adopting this perspective also leads to new conclusions about how and when theories of grammar competition can offer genuine explanations of patterns of change.

Bayesian inference involves formulating two or more hypotheses about the process by which the data were generated, and then comparing them in terms of their ability to explain the data. This approach has been argued to provide a solution to the over-/underfitting dilemma that plagues many approaches to statistical learning, both in statistics (Jeffreys 1961; MacKay 1992, 2003; Myung & Pitt 1997) and in cognitive science (Blanchard, Lombrozo & Nichols 2018; Xu & Tenenbaum 2007). The updated ("posterior") probability of each hypothesis, after being exposed to some evidence, is the product of (i) the hypothesis' probability before the evidence was received, and (ii) the probability that the hypothesis predicted for the observed evidence, in advance of seeing it. The second term—the "likelihood"—is a measure of the extent to which the hypothesis helps us to predict data that were actually observed, as opposed to other data that could have been observed but were not. This fact introduces a preference for hypotheses that make more specific predictions, as long as they are consistent with the evidence. Formally, for each hypothesis H_i out of a set of possible hypotheses $\mathcal{H}$, we have

$$P(H_i \mid D) = \frac{P(H_i) \times P(D \mid H_i)}{P(D)} = \frac{P(H_i) \times P(D \mid H_i)}{\sum_{H_j \in \mathcal{H}} P(H_j) \times P(D \mid H_j)}.$$

Often we write this equation up to proportionality, because the denominator is predictable if we know the full space of possible models.

$$P(H_i \mid D) \propto P(H_i) \times P(D \mid H_i)$$

For our purposes, the key property of Bayesian inference is the "Bayesian Ockham's Razor" (Jefferys & Berger 1991; MacKay 1992, 2003) or, as it is known in cognitive science, the "size principle" (Tenenbaum & Griffiths 2001). The posterior probability assigned to a hypothesis by Bayesian inference takes into account two sorts of questions. First, the prior: what beliefs did we already have about the relative probabilities of these models, before seeing the data? Second, the likelihood: how probable is it, according to each model, that the data would have been exactly as they were observed? This term depends on two conceptually distinct factors: first, whether the data *could in principle* have arisen in the form that they did. If not, the likelihood is zero, and so the posterior probability as well. If the data were in principle possible, then the likelihood term depends on how much probability the model would have put on the data before they were observed. When choosing between two models, both of which can accommodate the data, a Bayesian learner will prefer the more restrictive model: one that makes more specific predictions (has few parameters and little flexibility) rather than a less restrictive model that can in principle accommodate a greater variety of data. This feature of Bayesian learning thus offers a probabilistic version of the Subset Principle: see Perfors, Tenenbaum & Regier (2011) and Xu & Tenenbaum (2007) for discussion.

Note that the preference for simpler models in Bayesian inference is an effect of the likelihood term, not the prior. While some Bayesian models in statistics and cognitive science also impose a preference for simpler models in the form of the prior, this feature is independent of the simplicity preference under consideration and we do not use it here. The Bayesian Ockham's Razor effect arises even when the prior probability of all hypotheses is equal. In the latter case, the prior drops out and the relative probability of the two hypotheses is completely determined by their ability to predict the data:

$$P(H_i \mid D) = \frac{P(D \mid H_i)}{\sum_{H_j \in \mathbf{H}} P(D \mid H_j)},$$

or simply

$$P(H_i \mid D) \propto P(D \mid H_i).$$

Following Yang, we will assume that the learner initially assigns equal probability to all grammars, so that this is the form of the equation that is relevant to the probability learning in the grammar competition framework.

4.2 A hierarchical Bayesian learner for grammar competition

At first glance, a Bayesian model of probability learning for grammar competition might appear hopeless. If the hypotheses $H_i \in \mathcal{H}$ described above are identified with grammars, this is indeed the case. In a heterogeneous linguistic environment, there will generally be no grammar that can account for all of the data, with the result that a Bayesian learner who treats each grammar as a separate model would be forced to assign probability 0 to all grammars. The flaw in this model is that it implicitly assumes that the goal of inference is to make a guess about the single "true" grammar responsible for generating the data. There is no such grammar, since the data are produced by a statistical mixture of grammars.

The solution is to model the learner as one who knows that the linguistic environment is heterogeneous, and is actively trying to infer the statistical *mixture* of grammars that was responsible for producing the observed data. This is an instance of a **hierarchical Bayesian model**, a key concept in recent Bayesian statistics and cognitive science (see Gelman & Hill 2007; Kemp, Perfors & Tenenbaum 2007; Lee & Wagenmakers 2014 among many others). The idea is to model the data as having been generated by a multi-step process, each step of which is stochastic. Here, we assume that each data point (sentence) that a speaker produces is generated by selecting a grammar G_i according to a distribution over possible grammars $P(\mathbf{G})$. The choice of G_i, in its turn, determines the probability of generating data point (sentence) d. This happens with probability $P(d \mid G_i)$ according to facts about grammar G_i.[7]

The task of our Bayesian learner, then, is to infer the latent distribution $P(\mathbf{G})$ that is responsible for the characteristics of the observed data, from a space of candidate probability distributions. In the simplified case of two possible grammars G_1 and G_2 that we are focusing on, the distribution $P(\mathbf{G})$ is fully determined by a single parameter $q \in [0, 1]$ where $P(G_1) = q$ and $P(G_2) = 1 - q$. A Bayesian learner will make use of any available prior knowledge to form $P(q)$, but we assume here that the prior is uniform over all possible values: $P(q) \sim \mathcal{U}(0, 1)$.

This hierarchical model is depicted in Figure 5. Here, the parameter q only influences the distribution on grammars, which in turn fully determines the characteristics of the data D. In other words, the data are conditionally independent of q, given a choice of grammar. As a result, the learner can decompose the above equation into two probabilistic components corresponding to the arrows in Fig. 5: one for the choice of grammar, and another for the production of data given a grammar.

$$P(d \mid p) = P(d \mid G_1) \times P(G_1 \mid q) + P(d \mid G_2) \times P(G_2 \mid q)$$

By modeling language production as a multi-step generative process, each step of which may be stochastic, this hierarchical model efficiently encodes a higher-order probability distribution—a distribution over distributions (Pearl 1988). . This is the key to applying Bayesian learning in a grammar competition framework. A hierarchical learner does not simply assign a probability $P(G_i)$ to each grammar G_i, implicitly (and incorrectly) assuming that there is a true grammar responsible for the data. Instead, it uses the data to make inferences about the character of the probability distribution on grammars $P(\mathbf{G})$, on the sophisticated assumption that the data were generated from a variety of

[7]Notational conventions: As standard in probability theory, we use P ambiguously to talk both about distributions over random variables and about the probability that a variable takes on a specific value. So, for example, $P(\mathbf{G})$ is the distribution on all possible grammars, while $P(\mathbf{G} = G_1)$ is the probability that grammar G_1 is selected. When it is obvious from context which variable a value is selected from, the latter quantity is often abbreviated. So, references to $P(G_1)$ in the main text are short for $P(\mathbf{G} = G_1)$.

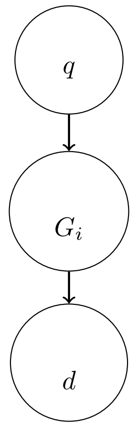

Figure 5: Hierarchical model of the relationship between the grammar mixture (q), the choice of grammar (G_i), and the observed characteristics of data points d.

distinct grammars. The learner's overarching goal is to infer what probabilistic mixture of grammars was responsible for the observed pattern of data. In the two-grammar case, any specific model is a hypothesis about the data-generating process of the form: "The data observed were generated by a mixture of grammars, G_1 with probability q and G_2 with probability $1-q$. Each grammar, in turn, is associated with a probability for each possible sentence."

In this approach, we can retain the structure of Yang's idealized model of learning and intergenerational transmission, modifying it only by replacing the LRP scheme with this learning rule: a hierarchical Bayesian learner, confronted with some data drawn from her linguistic environment, will update her theory of the latent distribution on grammars $P(\mathbf{G})$ by conditioning on the data received.

Using reasoning analogous to our discussion of the LRP above, consider again partially overlapping grammars G_1 and G_2 which generate sentences of type T_1—those parseable only by G_1—as well as sentences of type T_2, and sentences of type $T_{1/2}$ that can be parsed by either grammar.

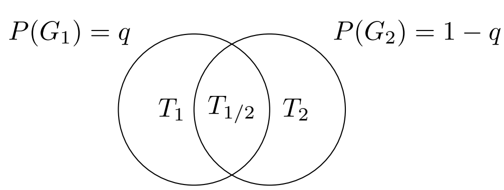

Figure 6: The linguistic environment created by grammars G_1 and G_2.

To compute the likelihood function for any data set, we find the probability that the learner will be presented with sentences of each type T_i, under a given hypothesis about the mixture q. Reasoning by cases, we can find the probability of a given data point d, relative to a hypothesized parameter q, as

$$\begin{aligned} P(d \in T_1 \mid q) &= q \times P(T_1 \mid G_1) \\ P(d \in T_2 \mid q) &= (1-q) \times P(T_2 \mid G_2) \\ P(d \in T_{1/2} \mid q) &= q \times P(T_{1/2} \mid G_1) + (1-q) \times P(T_{1/2} \mid G_2) \end{aligned}$$

Using the fact that each production of G_1 is either of type T_1 or $T_{1/2}$, and similarly for G_2, the likelihood function for data points in $T_{1/2}$ simplifies further to

$$P(d \in T_{1/2} \mid q) = q \times (1 - P(T_1 \mid G_1)) + (1-q) \times (1 - P(T_2 \mid G_2)).$$

In addition to the parameter q that we are trying to infer, this model has two unknowns $P(T_1 \mid G_1)$ and $P(T_2 \mid G_2)$, which we assume (following Yang) to be fixed features of the grammars (and/or their interaction with the sociolinguistic environment).

Maintaining Yang's idealization that the sentences presented to the learner are independent random draws from the environment E, the likelihood function for any data set D, for a hypothesized parameter q, is just the product of the likelihoods for each data point as given above.

$$P(D \mid q) = \prod_{d \in D} P(d \mid q)$$

Given this, we can compute the posterior probability of mixture parameter q—i.e., the probability of grammar G_1—given data set D, as the product of the likelihoods for each data point. Since each data point is either of type T_1, T_2, or $T_{1/2}$, the posterior is proportional to

$$\begin{aligned} P(p \mid D) &\propto P(D \mid p) && (12) \\ &= \prod_{T_i} \prod_{d \in D : d \in T_i} P(d \in T_i \mid p), && (13) \end{aligned}$$

where the T_i range over the three types under consideration, with likelihoods as given above.

4.3 Application to superset cases

Superset cases provide a revealing application. To model such an example, where the weak generative capacity of G_1 is strictly greater than that of G_2, all we have to do is to set $P(T_2 \mid G_2)$ equal to 0. Will the learning theory respond by assigning probability 1 to the more powerful grammar, as it did in the case of LRP? No, because the learning model does not only want to parse all utterances; it also wants to explain the statistical distribution of sentences of different types in the data, and doing so may require giving significant probability to the more restrictive grammar.

Consider the example of *pro*-drop, involving a competition between a grammar G_1, which can produce null-subject sentences, and G_2, which cannot. Sentences with null subjects are type T_1, and sentences with overt subjects are type $T_{1/2}$, parseable by both grammars. Because of the super-/subset relation there are no sentences of type T_2. Assume that the true value of q for the previous generation is .5, and that G_1 produces sentences with null subjects with some fixed probability—say, .4. Then the distribution of data in the environment will be, on average, 50% sentences produced by G_2 which both grammars can parse, and 50% sentences produced by G_1. Of the latter, 40% are unparseable by G_2 because they have null subjects—20% of all the data observed.

A data set of N sentences with these proportions will have on average $.2N$ sentences of type T_1. A generation of Bayesian learners provided with this data will then estimate q as

$$P(q \mid D) \propto \prod_{d \in D : d \in T_1} P(d \in T_1 \mid q) \prod_{d \in D : d \in T_{1/2}} P(d \in T_{1/2} \mid q)$$

This differs from the previous formula only in that the likelihood term for T_2-sentences does not occur, because there are no such sentences. Plugging in the values we are assuming for this example, the posterior distribution on q given a "typical" data set of size N—one with 20% null-subject sentences—is

$$\begin{aligned} P(q \mid D) \propto & \prod_{d \in D : d \in T_1} [q \times P(T_1 \mid G_1) + (1-q) \times P(T_1 \mid G_2)] \\ & \times \prod_{d \in D : d \in T_{1/2}} [q \times P(T_{1/2} \mid G_1) + (1-q) \times P(T_{1/2} \mid G_2)] \\ & = \prod_{d \in D : d \in T_1} [q \times .4 + 0] \prod_{d \in D : d \in T_{1/2}} [q \times .6 + (1-q) \times 1] \\ & = .4q^{.2N} \times [1 - .4q]^{.8N} \end{aligned}$$

This posterior is plotted in Figure 7 for a variety of data sets of size N. The most striking result is that the posterior is always centered on the true value of q that was involved in generating the data. As the number of observations N increases, the hierarchical Bayesian learner becomes increasingly confident of its estimate of this value (as evidenced by lower variance of the posterior distribution with larger N).

More generally, the posterior probability of each possible value of the mixture parameter q depends on how well the model could predict the distribution of data of all types, if q were the true value. That is, the inference process takes into account not only the ability of the various grammars to parse the sentences observed, but also the ability of the parameter q to explain the **number** of T_1 sentences observed. If q were too high, then the model would have difficulty explaining why sentences of Type T_1 do not occur more frequently in the data. In this context, the Bayesian Ockham's Razor effect appears in this way: the learner will prefer values of q that are high enough to ensure that T_1 sentences can be parsed, but not so high that we would expect more data points from the superset grammar than are actually observed.

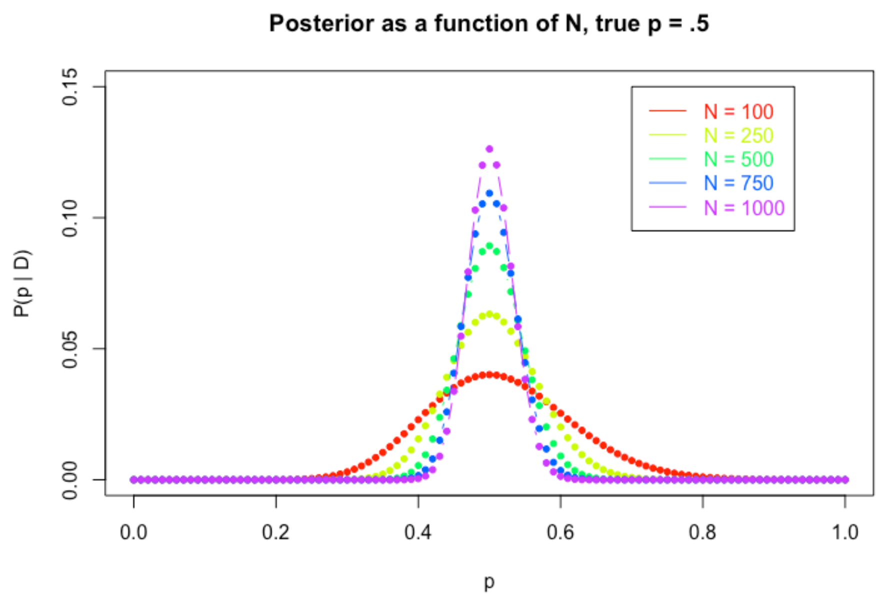

Figure 7: Grid approximation (size = 101) to the posterior distribution on parameters q, given various sizes of "typical" data sets with 20% null-subject sentences.

The result is, in the limit of infinite data, perfect learning of the true value of the mixture parameter. We can also observe this effect in Figure 8, where the posterior distribution is centred in each case on the true value of q (here, .5) that went into the previous generation's data-generating process. (The confidence of the posterior estimate does vary: this is because T_1-sentences are more informative than T_2 sentences, since they are unique to G_1. As a result, lower values of q that favor G_1 are associated with a greater proportion of T_1-sentences, leading to more informative data distributions.)

Unlike a reinforcement learner utilizing LRP, the hierarchical Bayesian learner induces a probability distribution over grammars by considering possible models of how the world is, and weighting them according to how well they explain the observed data. The Bayesian learner is unbiased: as long as the background model faithfully reflects the true generative process, the learner will infer a distribution $P(\mathbf{G})$ that is close to the true distribution that went into the generation of the data, with no tendency toward over- or underestimation on average. In short, the Superset Problem does not arise.

While we will not pursue the mathematical derivation in further detail, this result generalizes readily to the more general case with more than two grammars.

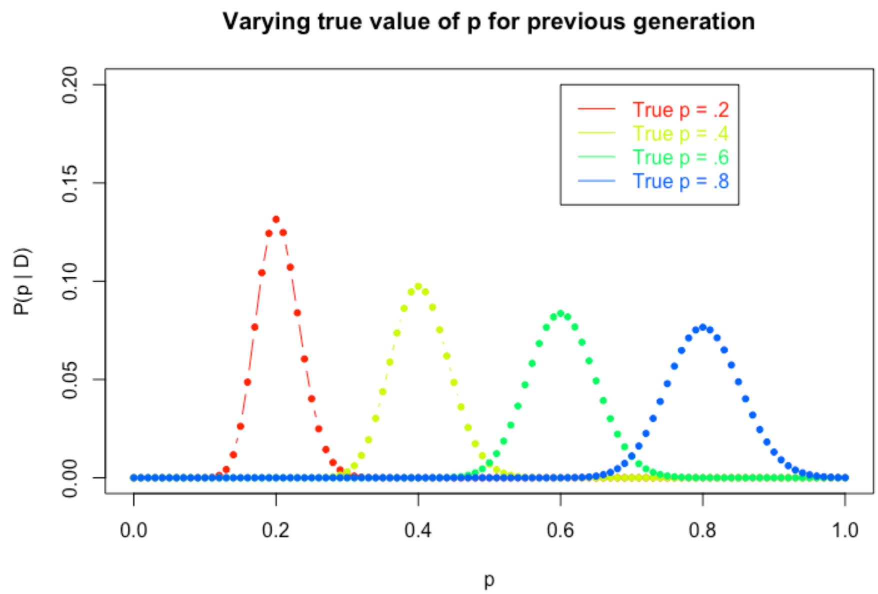

Figure 8: Learner's posterior distribution $P(p \mid D)$ for a range of different true values of the previous generation's mixture parameter q, assuming learners are exposed to "typical" data sets of size $N = 500$.

This also includes an expanded model which treats inference of continuous parameters controlling the values of $P(T_i \mid G_i)$ and the like (which can, if you like, be thought of as grammar competition in a continuous setting).

4.4 Diachronic predictions

The main result of the previous section is that the Superset Problem does not arise for the hierarchical Bayesian learner. With enough data the inferred value of q will be close to the true value, with no pressure toward higher or lower values regardless of the grammars' overlap in weak generative capacity.

As a result, the diachronic predictions of a theory built around a hierarchical Bayesian learner contrast starkly with the LRP-based theory. As we saw above, the dynamics of the LRP-based model are completely determined by $P(T_1|G_1)$ and $P(T_2|G_2)$—the frequency with which each grammar generates sentences that are not parseable by the other. Whichever grammar has a greater "advantage" will tend to take over, eventually obtaining probability 1. By contrast, these values do not play a major role in the process of Bayesian inference: instead, they merely provide further material for inferring the correct value of q, given statistical characteristics of the observed data.

This feature has important implications for the diachronic predictions of the

Andersen (1973)-inspired model of acquisition-driven change that Yang (2000; 2002) adopts. In a Bayesian Variational Model, each generation's inferred distribution over grammars $P(\mathbf{G})$ will match the previous generation's distribution, with differences attributable to statistical noise (i.e., drift). Successive generations of hierarchical Bayesian learners will tend toward stable co-existence of grammars, with no directional tendencies inherent to the learning model.

A model of syntactic change built around a hierarchical Bayesian learner avoids not just the Superset Problem, but also the problem of failed change. Specifically, the learning model does not generate any intrinsic pressures toward directionality in language change, regardless of linguistic features of the grammars involved. If some change has started and is on the rise at some point in time, this can be attributed either to statistical drift, or to non-linguistic (e.g., social) factors leading speakers to adopt one grammar/variant over others. Since the learning model is not responsible for the direction of the change, it is equally silent on the issue of whether the direction of the change can or should reverse. When a change-in-progress such as affirmative *do* reverses direction and heads toward extinction, this is equally compatible: it indicates that the external pressures that initially led to increase have changes, and now favor its decline.

5 Conclusion

This paper has proposed an acquisition-driven model of change within a grammar competition framework that is nearly identical in form to Yang's Variational Model, based on iterated learning. In this model, there are no intrinsic pressures for syntactic change in any direction. One important lesson of this discussion is that there is no intrinsic connection between the framework-level commitments of the Variational Model and the issues of directionality in syntactic change with which we began the paper. While the original version of the VM due to Yang (2000, 2002) does predict directional change in general and S-curves in particular, these predictions are attributable to the learning model assumed in that work. They are not attributable to the grammar competition model in general, and are only contingently related to the broader picture of acquisition-driven change associated with the Variational Model. As noted above, the choice to model statistical learning using LRP was not given any substantial motivation in Yang's original work, and it has not been systematically compared to alternative learning theories in the intervening decades. Furthermore, the LRP it is not employed in modern cognitive science research on language acquisition or other domains of learning.

This conclusion should come as a relief to theorists who are attracted to grammar competition framework or to a model of diachronic change based on iterated learning. The reason is that the prima facie attractive internal account of diachronic change associated with Yang's model and LRP turn out to hide a number of empirically false predictions: most crucially, the prediction that superset grammars always win in grammar competition, and that failed changes are impossible. There is, fortunately, a straightforward solution: we can retain the main framework-level features of Yang's theory and swap out LRP for a model of language learning that does not lead to pathological predictions about language change. Ideally, we could identify a theory that has substantial independent motivation from the literature on language acquisition and cognitive development more generally. While Bayesian learning is not the only model that meets these desiderata, it is well-motivated and fairly simple to apply in this context. We thus consider it a strong contender to replace LRP as the go-to learning theory in applications of the grammar competition model to language change.

The price, as noted above, is that a grammar competition theory modified in this way no longer generates theoretically interesting predictions about directionality in linguistic change. In particular, it does not predict the S-curve phenomenon. It is not clear whether this is a problem: as noted in the Introduction, many theorists have argued that directionality in language change is due to the interaction of linguistic and social factors, and cannot be explained merely by reference to internal properties of grammars (Blythe & Croft 2012; Labov 2001). In addition, S-curve patterns have been observed in the diffusion of a wide variety of cultural, social, and technological innovations (Rogers 1995). This supports our suggestion that an acquisition-based theory of grammatical change need not make any predictions about the directionality of change. Given that S-curves appear in the diffusion of innovations as diverse as corn-farming techniques and the stylistics of fake news, it is highly plausible that their appearance in linguistic change cannot be explained without reference to the social context in which change occurs.

It may be that a model such as the one proposed here, which does not attempt to account for the interaction of linguistic and social factors, is in principle unable to give a satisfying account of directionality in language change. At the least, our negative results with respect to the LRP-based theory suggest that efforts to do so should be treated with caution. However, there is considerable potential in the exploration of different learning theories within the broader grammar competition framework, including ones that might display gentler directional biases than Yang's does. In addition, several recent mathematical models of language change have formally treated social factors as a driving force in language change (Baxter et al. 2006; Guerrero Montero et al. 2023). However, to our knowledge, no existing formal model of language change attempts to treat systematically the interaction of linguistic and non-linguistic (e.g., social) factors. We believe that there is a great deal of potential in exploring this interaction in future work, within the grammar competition and other frameworks.

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